Holes are bonds: building molecules in 3D

molecules.fyi

KS3 chemistry. Use the free online model kit at **molecules.fyi** (no login needed). Every atom is a ball with holes, and **each hole is one bond**. A molecule is finished when every hole is filled.

Name: _____ Class: _____ Date: _____

How to use the kit

- Pick an atom from the box. Click the empty table to put it down, or click an atom to plug it in.
- **Fill with H** puts a hydrogen in every empty hole. **Undo** takes it back. **Clear** empties the table.
- **Link**: drag from one atom to another to join them. Do it again to make a double bond.
- The panel under the model shows the formula and the name of what you made.

1 Count the holes

Put one atom on an empty table and press **Fill with H**. Press **Clear** before the next one.

Atom	Holes	Name of what you made	Formula
Hydrogen		(skip)	(skip)
Oxygen			
Nitrogen			
Carbon			

Why can't an oxygen atom hold three hydrogens?

_____

2 The carbon staircase

Put down a carbon and press **Fill with H**. **Undo**, plug in another carbon, and fill again. Go up to five.

Carbons	1	2	3	4	5
Name					
Formula					

If a chain has *n* carbons, how many hydrogens does it have? _____

Predict the formula for 10 carbons: _____ Now build it to check.

3 Same formula, different molecule

Build two different molecules that are both C_4H_{10}: one a straight chain and one with a branch.

Straight chain: _____ With a branch: _____

The formula is the same. Why does the kit give them different names?

_____

4 Double bonds

Put down a carbon and plug a second carbon into it. Click **Link** and drag from one carbon to the other to add a second link. Click **Build**, then **Fill with H**.

Name: _____ Formula: _____

Drag the link in ethane (two carbons, one link) to twist it. Now try your double bond. What is different?

_____

Challenge: build carbon dioxide, CO_2, so that every hole is filled. How many links join each oxygen to the carbon? _____

Answers for teachers

1. Hydrogen 1 hole. Oxygen 2 holes: water, H_2O. Nitrogen 3 holes: ammonia, NH_3. Carbon 4 holes: methane, CH_4. Oxygen has only two holes, so it can make only two bonds.

2. 1: methane, CH_4. 2: ethane, C_2H_6. 3: propane, C_3H_8. 4: butane, C_4H_{10}. 5: pentane, C_5H_{12}. A chain of *n* carbons has $2n + 2$ hydrogens. 10 carbons: $C_{10}H_{22}$ (decane).

3. Butane and isobutane (its IUPAC name, 2-methylpropane, shows under it). The atoms are joined in a different order, so they are different substances: butane boils at about −1 °C and isobutane at about −12 °C.

4. Ethylene, C_2H_4 (the kit shows its IUPAC name, ethene, under it). The single bond in ethane twists. The double bond won't, and the kit says “Double and triple bonds can't twist”.

Challenge. Two links each: two double bonds, O=C=O.